

Acid-Base Balance MCQ – Chapter 40

Source: Short Textbook of Anesthesia – Ajay Yadav

Section 9: Cardiorespiratory Care (Chapter 40)

25 Questions • 15 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. In acute (noncompensated) respiratory acidosis, what happens to pH and pCO₂?

- A. pH normal, pCO₂ normal
- B. pH markedly increased, pCO₂ markedly decreased
- C. pH markedly decreased, pCO₂ markedly decreased
- D. pH markedly decreased, pCO₂ markedly increased

Ans: D

Q2. In chronic (compensated) respiratory acidosis, what is the expected HCO₃ level?

- A. Markedly increased
- B. Moderately increased
- C. Markedly decreased
- D. Normal

Ans: B

Q3. What is the pH finding in acute (noncompensated) respiratory alkalosis?

- A. Normal
- B. Moderately increased
- C. Mildly increased
- D. Markedly decreased

Ans: B

Q4. In chronic compensated respiratory alkalosis, what is the pH?

- A. Mildly increased or normal
- B. Markedly decreased
- C. Markedly increased
- D. Moderately decreased

Ans: A

Q5. In acute metabolic acidosis (noncompensated), what is the pCO₂ finding?

- A. Normal
- B. Mildly decreased
- C. Markedly decreased
- D. Markedly increased

Ans: B

Q6. Metabolic acidosis is defined as a decrease in pH below:

- A. 7.40
- B. 7.25
- C. 7.35
- D. 7.45

Ans: C

Q7. Which of the following is NOT a cause of metabolic acidosis?

- A. Diuretics
- B. Circulatory failure (shock)
- C. Renal failure
- D. Cyanide poisoning

Ans: A

Q8. The normal anion gap is: A. 8-16 mM/L (average 12 mM/L) B. 4-8 mM/L C. 2-6 mM/L D. 20-30 mM/L	Ans: A
Q9. The formula for anion gap is: A. Sodium – (Potassium + Chloride) B. Potassium – (Sodium + Chloride) C. Sodium – (Chloride + Bicarbonate concentration) D. Chloride – (Sodium + Bicarbonate)	Ans: C
Q10. Anion gap is mainly constituted by: A. Globulin B. Fibrinogen C. Hemoglobin D. Plasma albumin	Ans: D
Q11. According to Flow chart 40.1, if pH is decreased (<7.35) and pCO₂ is increased with increased HCO₃, the diagnosis is: A. Metabolic alkalosis B. Respiratory acidosis C. Respiratory alkalosis D. Metabolic acidosis	Ans: B
Q12. According to Flow chart 40.1, if pH is decreased (<7.35), pCO₂ is normal, and HCO₃ is decreased, the diagnosis is: A. Metabolic alkalosis B. Metabolic acidosis C. Respiratory alkalosis D. Respiratory acidosis	Ans: B
Q13. According to Flow chart 40.1, if pH >7.45, pCO₂ is decreased and HCO₃ is decreased, the diagnosis is: A. Respiratory acidosis B. Metabolic acidosis C. Metabolic alkalosis D. Respiratory alkalosis	Ans: D
Q14. According to Flow chart 40.1, if pH is normal (7.35-7.45), pCO₂ is increased, and HCO₃ is increased, the diagnosis is: A. Respiratory acidosis + Metabolic alkalosis B. Pure respiratory alkalosis C. Metabolic acidosis D. No acid-base abnormality	Ans: A

Q15. Which of the following is a cause of respiratory alkalosis? A. COPD B. Opioid overdose C. Myasthenia gravis D. Salicylate poisoning	Ans: D
Q16. Treatment of respiratory alkalosis includes: A. Infusion of ammonium chloride B. IV bicarbonate infusion C. Adjustment of ventilator setting (decrease frequency) and increasing rebreathing D. Increasing ventilator frequency	Ans: C
Q17. In COPD patients, respiratory acidosis may coexist with metabolic acidosis due to: A. Decreased cardiac output (cor pulmonale) and renal flow B. Excessive diuretic use C. Hyperventilation D. Excessive bicarbonate administration	Ans: A
Q18. Treatment of metabolic alkalosis includes IV infusion of: A. Normal saline only B. Potassium permanganate C. Sodium bicarbonate D. Ammonium chloride or 0.1N hydrochloric acid	Ans: D
Q19. The maximum rate of 0.1N HCl infusion for treating metabolic alkalosis should not exceed: A. 0.05 mEq/kg/hr B. 0.2 mEq/kg/hr C. 1.0 mEq/kg/hr D. 0.5 mEq/kg/hr	Ans: B
Q20. Which of the following is a cause of metabolic alkalosis? A. Renal failure B. Cyanide poisoning C. Hepatic failure D. Diuretics	Ans: D
Q21. In acute metabolic alkalosis (noncompensated), pCO₂ shows: A. Marked decrease B. Moderate increase C. Marked increase D. No change	Ans: B

Q22. According to Flow chart 40.1, if pH is decreased (<7.35), pCO₂ is increased, and HCO₃ is decreased or normal, the diagnosis is:

- A. Pure respiratory acidosis
- B. Metabolic alkalosis
- C. Pure metabolic acidosis
- D. Respiratory acidosis + Metabolic acidosis

Ans: D

Q23. Circulatory failure (shock) leading to accumulation of which acid causes metabolic acidosis?

- A. Lactic acid
- B. Uric acid
- C. Acetic acid
- D. Citric acid

Ans: A

Q24. According to Flow chart 40.1, if pH is normal (7.35-7.45), pCO₂ is decreased, and HCO₃ is decreased, the likely diagnoses include:

- A. Chronic metabolic acidosis (renal disease) and chronic respiratory alkalosis (pulmonary disease)
- B. Acute respiratory alkalosis
- C. No acid-base abnormality
- D. Respiratory acidosis + Metabolic alkalosis

Ans: A

Q25. In acute metabolic acidosis (noncompensated), HCO₃ shows:

- A. Marked decrease
- B. Mild decrease
- C. Moderate decrease
- D. No change

Ans: A

Answer Key & Explanations

Q1. Answer: D — pH markedly decreased, pCO₂ markedly increased

Page 294, Table 40.2: In acute respiratory acidosis (noncompensated), pH shows marked decrease (↓↓) and pCO₂ shows marked increase (↑↑↑).

Topic: Acid-Base Table

Q2. Answer: B — Moderately increased

Page 294, Table 40.2: In chronic compensated respiratory acidosis, HCO₃ shows moderate increase (↑↑) as renal compensation occurs.

Topic: Acid-Base Table

Q3. Answer: B — Moderately increased

Page 294, Table 40.2: In acute respiratory alkalosis (noncompensated), pH shows moderate increase (↑↑).

Topic: Acid-Base Table

Q4. Answer: A — Mildly increased or normal

Page 294, Table 40.2: In chronic compensated respiratory alkalosis, pH is mildly increased (↑) or normal.

Topic: Acid-Base Table

Q5. Answer: B — Mildly decreased

Page 294, Table 40.2: In acute metabolic acidosis (noncompensated), pCO₂ shows mild decrease (↓) as early respiratory compensation begins.

Topic: Acid-Base Table

Q6. Answer: C — 7.35

Page 294: Metabolic acidosis is defined as decrease in pH <7.35.

Topic: Metabolic Acidosis

Q7. Answer: A — Diuretics

Page 294: Causes of metabolic acidosis include renal failure, circulatory failure (shock) leading to accumulation of lactic acid, hepatic failure, diarrhea with loss of bicarbonate, and cyanide poisoning. Diuretics are a cause of metabolic alkalosis, not acidosis.

Topic: Metabolic Acidosis

Q8. Answer: A — 8-16 mM/L (average 12 mM/L)

Page 294: Normal anion gap = 8-16 mM/L (average 12 mM/L).

Topic: Anion Gap

Q9. Answer: C — Sodium – (Chloride + Bicarbonate concentration)

Page 294: Anion gap = Sodium concentration – (Chloride + Bicarbonate concentration).

Topic: Anion Gap

Q10. Answer: D — Plasma albumin

Page 294: Anion gap is mainly constituted by plasma albumin. It is also called unmeasured anion concentration, normally constituted by sulfates, phosphates, organic acids and plasma proteins.

Topic: Anion Gap

Q11. Answer: B — Respiratory acidosis

Page 294, Flow chart 40.1: pH decreased + increased pCO₂ + increased HCO₃ = Respiratory acidosis.

Topic: Diagnostic Approach

Q12. Answer: B — Metabolic acidosis

Page 294, Flow chart 40.1: pH decreased + normal pCO₂ + decreased HCO₃ = Metabolic acidosis.

Topic: Diagnostic Approach

Q13. Answer: D — Respiratory alkalosis

Page 294, Flow chart 40.1: pH increased (>7.45) + decreased pCO₂ + decreased HCO₃ = Respiratory alkalosis.

Topic: Diagnostic Approach

Q14. Answer: A — Respiratory acidosis + Metabolic alkalosis

Page 294, Flow chart 40.1: Normal pH + increased pCO₂ + increased HCO₃ = Respiratory acidosis + Metabolic alkalosis (mixed disorder with full compensation).

Topic: Diagnostic Approach

Q15. Answer: D — Salicylate poisoning

Page 294: Causes of respiratory alkalosis include Iatrogenic, Pregnancy, Salicylate poisoning, Hypoxia, and CNS trauma.

Topic: Respiratory Alkalosis

Q16. Answer: C — Adjustment of ventilator setting (decrease frequency) and increasing rebreathing

Page 294: Treatment of respiratory alkalosis: Adjustment of ventilator setting (decrease the frequency) and increasing the rebreathing (i.e. exhaled gases containing CO₂).

Topic: Respiratory Alkalosis

Q17. Answer: A — Decreased cardiac output (cor pulmonale) and renal flow

Page 294: Respiratory acidosis in COPD patients may be associated with metabolic acidosis due to decreased cardiac output (cor pulmonale) and renal flow.

Topic: Mixed Disorders

Q18. Answer: D — Ammonium chloride or 0.1N hydrochloric acid

Page 294: Treatment of metabolic alkalosis: Treat the underlying cause, IV infusion of ammonium chloride or 0.1N hydrochloric acid (not more than 0.2 mEq/kg/hr).

Topic: Metabolic Alkalosis

Q19. Answer: B — 0.2 mEq/kg/hr

Page 294: IV infusion of 0.1N hydrochloric acid should not be more than 0.2 mEq/kg/hr.

Topic: Metabolic Alkalosis

Q20. Answer: D — Diuretics

Page 294: Causes of metabolic alkalosis include Diuretics, Hypovolemia, Diarrhea with loss of chloride, and Iatrogenic.

Topic: Metabolic Alkalosis

Q21. Answer: B — Moderate increase

Page 294, Table 40.2: In acute metabolic alkalosis (noncompensated), $p\text{CO}_2$ shows moderate increase ($\uparrow\uparrow$) as respiratory compensation (hypoventilation) occurs.

Topic: Acid-Base Table

Q22. Answer: D — Respiratory acidosis + Metabolic acidosis

Page 294, Flow chart 40.1: pH decreased + increased $p\text{CO}_2$ + decreased/normal HCO_3^- = Respiratory acidosis + Metabolic acidosis (mixed disorder).

Topic: Diagnostic Approach

Q23. Answer: A — Lactic acid

Page 294: Causes of metabolic acidosis include circulatory failure (shock) leading to accumulation of lactic acid.

Topic: Metabolic Acidosis

Q24. Answer: A — Chronic metabolic acidosis (renal disease) and chronic respiratory alkalosis (pulmonary disease)

Page 294, Flow chart 40.1: Normal pH + decreased $p\text{CO}_2$ + decreased HCO_3^- = Chronic metabolic acidosis (due to renal disease) and chronic respiratory alkalosis due to pulmonary disease.

Topic: Diagnostic Approach

Q25. Answer: A — Marked decrease

Page 294, Table 40.2: In acute metabolic acidosis (noncompensated), HCO_3^- shows marked decrease ($\downarrow\downarrow\downarrow$).

Topic: Acid-Base Table